MIL-G-43196C
12 March 1987
SUPERSEDING
MIL-G-43196B
28 November 1978

MILITARY SPECIFICATION

GLOVES, ROCKET FUEL HANDLERS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers impermeable coated gloves with closure ring for wear with special protective clothing for the handling of liquid fuels and oxidizers (except for liquid oxygen and nitrogen).

1.2 Classification. The gloves shall be one type in the following sizes (see 6.2):

Schedule of sizes

Small
Medium
Large
Extra-large

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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SPECIFICATIONS

FEDERAL

V-T-276	- Thread, Cotton
ZZ-G-381	- Gloves, Rubber, Industrial
MMM-A-189	- Adhesive, Synthetic-Rubber Thermoplastic, General-Purpose
PPP-B-566	- Boxes, Folding, Paperboard
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-B-665	- Boxes: Paperboard, Metal Edged and Components
PPP-B-676	- Boxes, Setup
PPP-T-45	- Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

MILITARY

MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-P-7254	- Propellant, Nitric Acid
MIL-L-35078	- Loads, Unit: Preparation of Semiperishable Subsistence Items, Clothing, Personal Equipment and Equipage; General Specification For
MIL-T-43548	- Thread, Polyester, Cotton-Covered and Rayon Covered

STANDARDS

FEDERAL

FED-STD-191	- Textile Test Methods
FED-STD-595	- Colors
FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
MIL-STD-731	- Quality of Wood Members for Containers and Pallets

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DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

2-1-455	- Glove Dipping Forms, Illustrative Views
8-1-2	- Jig, Acid Testing, No. 1

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 413	- Rubber Property - Adhesion to Flexible Substrate
D 638	- Tensile Properties of Plastics
D 648	- Deflection Temperature of Plastics Under Flexural Load
D 792	- Specific Gravity and Density of Plastics by Displacement
D 3951	- Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Guide samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variations from the document may appear in the sample, in which case this document shall govern.

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3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.3 Materials. (See 6.5).

3.3.1 Base fabric. The base fabric for coating shall be an unbleached, knitted, cotton cloth made with an interlock stitch. One side of the fabric shall be lightly brushed or napped to give a fleeced effect. The base fabric shall meet the requirements of table I when tested as specified in 4.4.1.1.

TABLE I. Base fabric

Characteristic	Requirement (minimum)
Weight (ounces per square yard)	5.8
Wales per inch	36
Courses per inch	28
Bursting strength (pounds)	90

3.3.2 Closure ring. The closure ring shall be molded from natural color, virgin, unfilled polypropylene. The molded pieces shall be uniform in color, texture, and finish and shall be free from cracks, blisters, scratches, or other defects affecting serviceability. The closure ring shall conform to the requirements shown in table II when tested as specified in 4.4.1.1 and shall conform to the dimensions of figure 1.

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TABLE II. Closure ring properties

Characteristics of polypropylene	Requirements Minimum	Maximum
Specific gravity	0.895	0.904
Tensile strength (psi)	4,000	-
Deflection temperature (°F)	190	-

3.3.3 Coating compounds.

3.3.3.1 Base dipping compound. The dipping compound shall be a virgin polyvinyl chloride or a vinyl chloride, acetate copolymer containing not more than 4 percent polyvinyl acetate, plasticized with phosphate of phthalate ester plasticizers exclusively, and colored gray (see 3.8).

3.3.3.2 Final dip coating compound. The final dip coating shall contain unplasticized particles of powdered vinyl resin capable of imparting a roughened texture to the coated gloves.

3.3.4 Coated gloves. The coated fabric in the finished gloves shall conform to the requirements of table III when tested as specified in 4.4.2.2 and 4.4.5.

TABLE III. Properties of the finished gloves

Test	Requirements Minimum	Maximum
Breaking strength in wale direction (pounds)	50	-
Stiffness (cm):		
At 73.5°F in wale direction	-	11.0
At 0°F in wale direction	-	12.0

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TABLE III. Properties of the finished gloves - Continued

Test	Requirements Minimum	Maximum
Blocking	-	No greater than scale rating No. 2
Adhesion of coating (pounds per 2 inches)	6.0	-
Thickness of coating (inches)	0.025	0.040
Nitric acid resistance 1/	Pass	-
JP4 fuel resistance 2/	Pass	-
Porosity test	Pass	-
Adhesion of glove to closure ring (pounds per inch)	1.0	-
Crocking of marking 3/:		
Dry	Good	-
Wet	Good	-

* 1/ There shall be no evidence of cracking, flaking, separation of the coating from the fabric, or change in color of the indicator solution. The test specimen shall not show more than an appreciable change in stiffness.

* 2/ There shall be no evidence of cracking, softening, flaking, or separation of coating from the fabric. The test specimen shall not show more than an appreciable change in stiffness.

3/ The marking shall have good legibility when tested as specified in 4.4.5.

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* 3.3.5 Thread. The thread for sewing and stitching the gloves shall be unbleached cotton thread conforming to type IA1 of V-T-276. The needle thread shall be no finer than 50/3 and the looper thread shall be no finer than 70/3. As an alternative, covered polyester thread conforming to MIL-T-43548 in equivalent 2 or 3 ply counts may be used.

3.4 Design. The gloves shall be a curved five finger style, with closure ring and cotton inner shell adhered to an outer vinyl layer made over dipping forms conforming to the guide forms furnished by the Government (see 3.7.1).

3.5 Guide patterns. The working patterns used for cutting the base fabric for the glove shells shall be based upon the guide patterns furnished by the Government. Government furnished guide patterns do not provide for necessary seam allowances. The pattern parts and cutting quantities shall be as follows:

Pattern parts	Cutting quantities per pair
Palm side of shell including front of thumb, front, and side portion of all fingers	2
Back side of shell including back of thumb and side portion of all fingers	2

3.6 Construction.

3.6.1 Cutting. The component parts of the cloth glove shells shall be cut with the length dimension (palm to fingertip direction) in the length direction (wales) of the fabric.

3.6.2 Seams, stitches, and stitching of the gloves. The seams shall conform to type SSa-1 and the stitches to type 301 or 401 of FED-STD-751. All seaming and stitching shall be done with not less than 10 nor more than 14 stitches per inch. There shall be not less than 1/16 inch nor more than 3/16 inch seam allowance on all seams. Sewing machines with trimmer attachments may be used.

3.6.2.1 Type 301 stitches. Ends of stitching shall be backstitched or overstitched not less than 3/8 inch except where ends are caught in other seams and stitching. Thread tension shall be maintained so there will be no loose stitching resulting in loose bobbin or top thread, or excessively tight stitching resulting in puckering of the material sewn. The lock shall be embedded in the materials sewn.

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3.6.2.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-outs occur during sewing, the stitching shall be repaired by restarting the stitching a minimum of 3/8 inch back of the end of the stitching.

b. Thread breaks or two or more consecutive skipped or run-off stitches noted during inspection of the item (in-process or end item) shall be repaired by overstitching. The stitching shall start a minimum of 3/8 inch back of the defective area, continue over the defective area, and continue a minimum of 3/8 inch beyond the defective area onto the existing stitching. Loose or tight stitching shall be repaired by removing the defective stitching without damaging the material and restitching in the required manner. 1/

1/ When making the above repairs, the ends of stitching are not required to be backstitched.

3.6.2.2 Type 401 stitching. Thread tension shall be maintained so there will be no loose stitching. Where stitch type 401 is used, the looper thread shall be on the inside of the gloves. Both ends of all seam or stitching produced with 401 stitch type, when not caught in other seams or stitching, shall have a 1/2 to 3/4 inch chain extending beyond each end. All repairs shall be in accordance with 3.6.2.1.1.a and 3.6.2.1.1.b.

3.6.3 Turning of the cloth glove shells. The cloth glove shells shall be turned so that the seams and fleeced effect shall be on the inside of the completed glove.

3.6.4 Mounting of cloth glove shells. The turned cloth glove shells shall be mounted on forms conforming to 3.7.1. Seams shall be positioned at the sides and pulled back in the fingertip and thumb-tip areas.

3.6.5 Coating of cloth gloves. The mounted cloth gloves shall be evenly and completely dipped in the base dipping compound specified in 3.3.3.1. The rough finish imparted by the final dip coating specified in 3.3.3.2 shall extend in length from the tips of the fingers and thumb toward the cuff portion to a point 3 to 4 inches from cuff edge. After application of coating, the compounds shall be fully fused (individually or simultaneously). There shall be no separation between coatings, and there shall be no more strike-through than that exhibited by the standard sample (see 6.4).

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3.6.6 Placing and adhesion of closure rings. The closure rings specified in 3.3.2 shall be inserted into the gloves, beaded end first, and positioned in the space provided by the dipping forms. The top edge of the gloves shall be even with the top edge of the rings. The closure rings shall be securely bonded to the gloves with an adhesive conforming to MMM-A-189 (see 6.6).

3.7 Dimensions.

3.7.1 Dipping forms. The dimensions of the dipping forms for these gloves shall be in accordance with the measurements shown on Drawing 2-1-455. Production forms may be modified to include a removable thumb in lieu of a permanent thumb providing the dimensions shown in Drawing 2-1-455 are met. Dipping forms shall conform to guide forms furnished by the Government (see 3.4).

* 3.7.2 Finished gloves. The finished glove dimensions shall conform to the requirements shown in table IV when measured as specified in 4.4.4 (see figure 2).

TABLE IV. Dimensions of finished gloves

Size	Length (A) inches ± 1/4 inch	Width across palm (B) inches ± 1/16 inch
Small	12 1/2	4 1/2
Medium	13	4 3/4
Large	13 1/2	5
X-Large	14	5 1/2

3.7.3 Closure ring. The finished dimensions of the closure ring shall conform to the measurements shown in figure 1.

* 3.8 Color. The color of the glove coating compound shall approximate Gray 16187 of FED-STD-595.

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3.9 Marking.

3.9.1 Instruction marking for gloves. Each glove shall be permanently and legibly marked in letters not less than 1/4 inch in height with a black marking medium on the outside back of the gauntlet (centered approximately 1 inch from the top and parallel to the top open edge) with the following legend:

"DO NOT USE FOR ELECTRICAL WORK"

* 3.9.2 Identification marking for gloves. Each glove shall be permanently and legibly marked with a black marking medium on the outside (palm side) of the gauntlet (centered approximately 1/2 inch from the top and parallel to the top open edge) in letters not less than 1/4 inch in height with the following information:

GLOVES, ROCKET FUEL HANDLERS (EXCEPT LIQUID OXYGEN AND NITROGEN)

CONTRACT NUMBER
STOCK NUMBER
SIZE

* 3.10 Instruction sheet. A paper instruction sheet shall be placed inside one glove of each pair of gloves. The legend shall be legibly printed or typed in black characters. The instruction sheet shall be white paper with the following legend:

GLOVES, ROCKET FUEL HANDLERS

1. For protection against injury from oxidizers and liquid fuels.
2. These gloves are to be worn only when wearing either limited or full protection suit.
3. If oxidizers or liquid fuels come in contact with gloves, wash them in water before removing.
4. If gloves become torn, or if protective coating is cracked or damaged, discard gloves immediately and use a serviceable pair.
5. To attach gloves to suit, roll up gum wristlet on sleeve of protective suit. Slip closure ring attached to glove over the closure ring attached to sleeve of suit. Roll gum wristlet of suit down over glove closure ring.
6. WARNING. Not to be used with liquified gases that are maintained or handled at extremely low temperatures, such as liquid oxygen, nitrogen, etc.

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3.11 Replacement of defective components. During the manufacturing process, components having material defects or damages that are classified as defects in 4.4.2.1, 4.4.3, and 4.4.4 shall be removed from production and replaced with nondefective and properly matched components.

3.12 Workmanship. The finished gloves shall conform to the quality of product established by this document. Occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4. The presence of any defect shall be cause for rejection of the first article.

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4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

4.4.1.1 Component testing. Inspection shall be performed on components and materials listed in table V for characteristics shown. All test requirements shall be applicable to the individual unit. For each component, the inspection level shall be S-1 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 4.0.

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TABLE V. Component tests

Component and lot size expressed in terms of	Characteristic	Requirement para.	Test Method	No. determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically to nearest	Sample unit
Base fabric (1 yard full width)	Material identification	3.3.1	1200 1/	1	X	-	
	Knitted stitch	3.3.1	Visual	1	X	-	
	Weight	Table I	5041 1/	5	-	0.1 oz per sq yd	1/2 yd
	Wales	Table I	5070 1/	5		Wale per in.	
	Courses	Table I	5070 1/	5		Course per in.	
	Bursting strength	Table I	5120 1/	5		1 lb	
Glove closure ring (1 each)	Identification (composition)	3.3.2	2/	1	X	-	
	Specific gravity	Table II	ASTM D 792	1		0.001	
	Tensile strength	Table II	ASTM D 638	1		10 psi	Two closure rings
	Deflection temperature	Table II	ASTM D 648 3/	1		1°F	

1/ Test method of FED-STD-191.
2/ A certificate of compliance shall be submitted and will be accepted for the stated requirements.
3/ The specimen shall be subjected to a maximum fiber stress of 66 psi. ,

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4.4.2 In-process inspection.

4.4.2.1 Visual examination of cloth inner glove shells. The cloth inner glove shells shall be examined for stitching and seaming defects in table VI. The lot size shall be expressed in units of one glove shell. The sample unit shall be one glove shell. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 1.0 for major defects and 2.5 for total defects. Any critical defect found during the sampling inspection will be a cause for rejection of the lot. One hundred percent inspection shall be performed for critical defects on each lot found acceptable under the sampling inspection. Any critical defect found during the 100 percent examination shall result in rejection of the glove shell containing the critical defect. The examination for all stitching and seam defects except for open seams and seams on outside shall be made on glove shells taken prior to turning. Fxamination (100 pecent) for open seams and for seams on outside shall be made with glove shells turned and stretched on appropriate dipping forms prior to dipping operation. A written report covering each in-process lot shall be required.

TABLE VI. Cloth inner glove shell defects

Examine	Defect	Classification		
		Critical	Major	Minor
Cotton cloth	Any hole, cut, or tear	X		
Seams and stitching	Seam type not as specified		X	
	Stitch type not as specified		X	
	Repair of seam not as specified			X
	Open seam	X		
NOTE:	A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or run-off stitches occur.			

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TABLE VI. Cloth inner glove shell defects - Continued

Examine	Defect	Classification		
		Critical	Major	Minor
Seams and stitching - continued	Stitching omitted where required	X		
	Two or more stitches per inch less than specified		X	
	One less than specified stitches per inch			X
	More than maximum number of specified stitches per inch, but not damaging the fabric			X
	Seams not on inside of glove	X		
	Tight tension resulting in puckering of seam		X	
	Less than 1/16 inch seam allowance		X	
	More than 3/16 inch seam allowance			X
	Tight tension causing noticeable puckering and thread breaks when seam is pulled	X		
	Thread not as specified		X	
	Ends of stitching not secured		X	

4.4.2.2 Testing of gloves before adding closure ring. The coated gloves (before adding the ring closure) shall be tested for the characteristics listed in table VII. The methods of testing specified in FED-STD-191 wherever applicable and as specified in table VII shall be followed. The lot size shall be expressed in units of gloves. The sample unit shall be ten gloves: five right-handed gloves and five left-handed gloves. The requirements for all tests are applicable to the sample unit. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5, except for those test characteristics that have an asterisk (*). The test characteristics designated by an asterisk (*) shall be considered critical, and any test failure shall result in rejection of the lot represented by the sample.

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TABLE VII. Glove tests (before adding closure ring)

Characteristic	Requirement paragraph	Test method	No. determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically nearest to
Breaking strength in wale direction	3.3.4	5100 1/	5	-	1 lb
Blocking	3.3.4	5872 1/ 2/	3	X	-
Stiffness at 73.5°F in wale direction	3.3.4	5204 1/ 3/	5	-	0.1 cm
Stiffness at 0°F in wale direction	3.3.4	5204 1/ 3/ 4/	5	-	0.1 cm
Adhesion of coating	3.3.4	4.5.1	3	-	0.1 lb
Thickness of coating	3.3.4	4.5.2	5	-	0.001 inch
Nitric acid resistance*	3.3.4	4.5.3	1	X	-
JP-4 fuel resistance*	3.3.4	4.5.4	1	X	-
Porosity test*	3.3.4	4.5.5	6	X	-

1/ Test method is specified in FED-STD-191.
2/ The specimen shall be 2 by 4 inches. The glass plates shall be 2 1/2 inches by 2 1/2 inches by 1/8 inch. The weight shall be 2 pounds. Evaluation shall be in accordance with scale 2 of referenced test method.
3/ Specimens shall be taken from the cuff area of the gloves with the long dimension of the specimen in the wale direction of the fabric. Care shall be taken to ensure the specimen includes as much of the smooth area (unroughened area) as is possible. The end of the specimen that has the roughened texture shall be placed in the nip rolls of the test apparatus.
4/ Specimens shall be conditioned at a temperature of 0°F ± 2°F for 4 hours. At the end of the 4 hours conditioning period, the specimens shall be immediately tested for stiffness in a still atmosphere and at a temperature of 0°F ± 2°F.

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* 4.4.3 Critical defect examination of end item. Prior to performing the end item sampling examination required in 4.4.4, the gloves shall be 100 percent examined for the critical defects listed in table VIII. This 100 percent critical defect examination shall become a part of the contractor's inspection system or quality program. Any glove found to contain a critical defect shall be rejected.

* 4.4.4 End item visual examination. The end item shall be examined for the defects listed in table VIII. The lot size shall be expressed in units of gloves. The sample unit shall be one glove and selection shall be by pairs. The sample unit for pairing defects shall be one pair. The inspection level shall be II and the AQL, expressed in terms of defects per hundred units, shall be 2.5 for major defects and 6.5 for total combined major and minor defects. Any critical defect found during sampling inspection shall result in rejection of the lot.

TABLE VIII. End item visual defects

Examine	Defect	Classification		
		Critical	Major	Minor
Pairing	Definite variation in color or appearance			X
	Gloves not mated in pairs (e.g., two left or two right gloves submitted as a pair or not same size)		X	
Cleanness	Any spot, stain, or foreign matter that requires a cleaning agent for removal			X
Color of glove	Not gray		X	
Color of closure ring	Not, natural			X

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TABLE VIII. End item visual defects - Continued

Examine	Defect	Classification		
		Critical	Major	Minor
Design	Any characteristic not in accordance with the specified requirements:			
	-seriously affecting serviceability		X	
	-affecting serviceability but not seriously			X
Coated fabric	Any hole, cut, abraded area, tear, or rip through material or exposing base fabric	X		
	Exterior roughened texture omitted		X	
	Roughened texture extending from tips of fingers and thumbs to a point:			
	-more than 6 inches down from cuff edge		X	
	-less than 3 inches down from cuff edge			X
	-more than 4 inches but not more than 6 inches down from cuff edge			X
	Any evidence of separation of coating		X	
	Area of no coating	X		
	Any brittle area that when bent results in a crack	X		
	Area of excessive coating (i.e., solid run(s), ridge(s) or lump(s):			
	-seriously affecting serviceability		X	
	-affecting serviceability, but not seriously			X

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TABLE VIII. End item visual defects - Continued

Examine	Defect	Classification		
		Critical	Major	Minor
Coated fabric - continued	Any pit or abraded area but not through material or not to a degree where base fabric is exposed		X	
	Any embedded foreign material that when removed will expose base fabric or leave a through hole	X		
	Foreign inclusion that cannot be readily removed or if removable does not expose base fabric:			
	-seriously affecting serviceability		X	
	-affecting serviceability but not seriously			X
Construction and workmanship	Not constructed as specified:			
	-seriously affecting serviceability		X	
	-affecting serviceability but not seriously			X
	Any component or assembly malformed:			
	-seriously affecting serviceability		X	
	-affecting serviceability but not seriously			X
	Any component damaged, or misplaced or operation omitted or not properly performed:			
	-seriously affecting serviceability		X	
	-affecting serviceability but not seriously			X

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TABLE VIII. End item visual defects - Continued

Examine	Defect	Classification		
		Critical	Major	Minor
Ring closure	Omitted	X		
	Sharp edges not removed from ends		X	
	Any blister, crack, fracture, or other defect:			
	-seriously affecting serviceability		X	
	-affecting serviceability but not seriously			X
	Not attached as specified:			
	-seriously affecting serviceability		X	
	-affecting serviceability but not seriously			X
	Top edge of glove not even with unbeaded edge of closure ring:			
	-by more than 1/4 inch		X	
	-by more than 1/16 inch but not more than 1/4 inch			X
Markings and instruction slip	Not in specified location, characters not of height specified, not specified color, or not accomplished as required			X
	Omitted or illegible		X	
	Incomplete or incorrect to the extent that the intended purpose will not be served		X	

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* 4.4.5 End item dimensional examination. The gloves and closure rings shall be examined for conformance to the dimensions specified in table IV and figure 1, respectively. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of completely fabricated gloves. The sample unit shall be one glove with closure ring. For overall length (A), the measurement shall be made on the back, from the tip of the middle finger in the curved position to the top open edge of glove. For width across palm (B), the glove shall be flattened without stretching and the width measured on a line passing through the thumb and crotch and parallel to open top edge of the glove (see figure 2). The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

4.4.6 End item testing. The end item shall be tested in accordance with table IX. The lot size shall be expressed in units of one glove. The sample unit shall be two complete gloves, selected by pairs. The requirements are applicable to the sample unit. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

TABLE IX. End item tests

Characteristic	Requirement paragraph	Test method	No. determinations per sample unit	Results reported as: Pass or fail	Results reported as: Numerically nearest to
Adhesion of closure ring to glove	3.3.4	4.5.6	1	-	0.1 lb/inch
Crocking of marking					
Dry	3.3.4	4.5.7	1	X	-
Wet	3.3.4	4.5.7	1	X	-

4.4.7 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling Bulged or distorted container
Content	Number of unit packs per container is more or less than required 1/

1/ For this defect, one shipping container in the sample shall be examined.

* 4.4.8 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

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4.5 Methods of inspection.

4.5.1 Adhesion of coating test. Adhesion of coating shall be determined in accordance with Method 5970 of FED-STD-191 with the following exceptions:

a. Area of the glove to be tested shall be the unroughened area.
b. Test specimen shall be prepared by adhering two plies together with a suitable adhesive (see 6.7).
c. The provisions of Method 5970 requiring that the coating separate completely from the cloth for the entire 2 inch width shall not apply if the plies of the specimen separate without removing the coating, provided the recorded resistance to separation of the plies is greater than the requirement specified for adhesion.

4.5.2 Thickness of coating test. The coating shall be stripped from the base fabric. (Sample may be soaked in methyl ethyl ketone (MEK) to facilitate separation of coating from base fabric.) The gage used for the measurement of thickness shall be of a dead-weight type equipped with a dial graduated to read directly to 0.001 inch. The presser foot shall be circular with a diameter of 0.25 ± 0.01 inch. The presser foot and moving parts connected therewith shall be weighted so as to apply a total load of 3 ounces ± 0.1 ounce to the specimen. The presser foot and anvil surface shall be planed to within 0.0001 inch and parallel to each other within 0.0001 inch. The thickness shall be measured on the smooth surface area of the glove rather than the area covered by the rough final dip coating.

4.5.3 Nitric acid resistance test. Two 1 3/8 inch diameter specimens shall be die cut, one each from roughened (hand) portion and unroughened (gauntlet) portion of the glove to be tested. Each specimen shall be placed in the test jig shown on Drawing 8-1-2 with the outside of the glove in contact with the acid. Approximately two drops of the indicator solution shall be placed on a supporting glass plate, and the bottom surface of the test assembly sealed to the plate with a petroleum jelly seal. A mirror shall be so situated that any change in color of the indicator may be observed. Approximately 20 mL of fuming nitric acid conforming to type IIIA of MIL-P-7254 shall be placed in the jig. The test jig shall be covered with a glass plate with a petroleum jelly seal between the glass cover and the top rim of the jig. Penetration time shall be taken as the time from acid contact with the test specimen to the first visual acidic color change of the indicator. Indicator solution shall be 0.12 percent Congo red dye in distilled water by weight. After 8 minutes of exposure to acid, disassemble test jig, wash test specimen, dry, and examine for the properties in table III. Specimens from both roughened and unroughened surface of glove shall pass acid resistance requirements of table III.

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4.5.4 JP-4 resistance test. Two 1 3/8 inch diameter specimens shall be cut, one from roughened (hand) portion and unroughened (gauntlet) portion of the glove to be tested. Each specimen shall be placed in the test jig shown on Drawing 8-1-2 with the coated side in contact with the fuel. Approximately 20 mL of JP-4 conforming to MIL-T-5624 shall be placed in the test jig. The specified fuel shall contact the test specimen for 15 minutes, after which the test sample shall be removed from the test jig and dried at room temperature for 4 hours. When dry, the test specimen shall be folded in half across the diameter of the test specimen and rolled with a roller. The roller shall weigh approximately 10 pounds. The test specimen shall be folded again across the diameter perpendicular to the direction of the first fold and rolled again in the same manner. Each specimen shall then be examined visually for evidence of cracking, softening, stiffening, flaking, or separation of coating from the fabric.

4.5.5 Porosity test. The porosity test shall be performed as specified in ZZ-G-381.

4.5.6 Adhesion test of closure ring to glove. Adhesion of closure ring to gloves shall be determined as specified in ASTM D 413. Resistance to separation of the coated fabric from the ring shall be determined and recorded on the automatic recording chart for a minimum of 1 inch of separation.

4.5.7 Crocking of marking test. Crocking, dry and wet, of marking shall be determined as specified in Method 5651 of FED-STD-191 except that the "Evaluation" specified in Method 5651 shall not apply. Evaluation of crocking, both dry and wet, shall be based on legibility of marking after test. Samples for test shall be carefully peeled from the glove closure ring. The sample shall then be mounted in the test apparatus in such a manner that the cylindrical finger of the test apparatus passes over the full length of the line of marking in its movement cycle. Crocking of marking shall be visually rated after test as follows:

Good - No appreciable loss of legibility.

NOTE: The type of crock cloth used shall be reported.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. Each pair of gloves, alternated end for end with palms facing inward, shall be unit packed on edge in a folding paperboard box conforming to variety 1, style III, type B, class a of PPP-B-566; setup paperboard box conforming to type I, variety 1, class A, style 4 of PPP-B-676;

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or metal-edged paperboard box conforming to style A of PPP-B-665. The inside dimensions of the paperboard box shall be 12 inches in length, 6 1/2 inches in width, and 5 inches in depth. Box closure shall be effected with 2-inch minimum width gummed paper tape conforming to type III, grade B of PPP-T-45, applied at the center of the length opening and extending along the bottom and up the sides at least 2 inches.

* 5.1.2 Commercial. Gloves shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

* 5.2.1 Level A packing. Twenty-four pairs of gloves of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, grade V2s of PPP-B-636. Level A unit packs shall be packed flat six in length, two in width, and two in depth. Inside dimensions of each shipping container shall be approximately 41 inches in length, 24 1/2 inches in width, and 10 1/2 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.7. Toward the end of the contract or when there is less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Twenty-four pairs of gloves of one size only, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A unit packs shall be packed flat, six in length, two in width, and two in depth. Inside dimensions of each shipping container shall be approximately 41 inches in length, 24 1/2 inches in width, and 10 1/2 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.4.7. Toward the end of the contract or when there is less than the required amount per container of the same size, mixed sizes may be packed within the same shipping container.

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5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the shipping container shall be grade V3c, V3s, or V4s, fabricated in accordance with PPP-B-636, and closed in accordance with method III as specified in the appendix of PPP-B-636, except that inspection shall be in accordance with 4.4.7.

* 5.2.3 Commercial. Gloves, unit packed as specified in 5.1, shall be packed in accordance with ASTM D 3951.

* 5.3 Palletization. When specified (see 6.2), gloves, packed as specified in 5.2.2 and 5.2.3, shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 89 in accordance with appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

* 5.4 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers, and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. These gloves are intended for use with the Standard A limited and full protective rocket fuel handlers' clothing. They may also be used with the Standard B limited and full protective rocket fuel handlers' clothing.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Sizes required (see 1.2).
c. When a first article is required (see 3.2, 4.3, and 6.4).
d. Selection of the applicable levels of preservation and packing (see 5.1 and 5.2).
e. Type and class of unit load required (see 5.2.1).
f. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
g. When palletization is required (see 5.3).

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* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Samples. For access to samples, address the contracting activity issuing the invitation for bids.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.3).

6.6 Adhesion of closure ring. It has been found that closure ring adhesion is satisfactory when the closure rings are first roughened and immersed in a saturated solution of chromic acid in concentrated sulphuric adhesive.

6.7 Adhesive. It has been found that adhesive 3M No. 776 is adequate for use in preparing adhesion of coating samples.

* 6.8 Subject term (key word) listing.

Gloves
Rocket fuel handlers
Protective clothing

6.9 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.)

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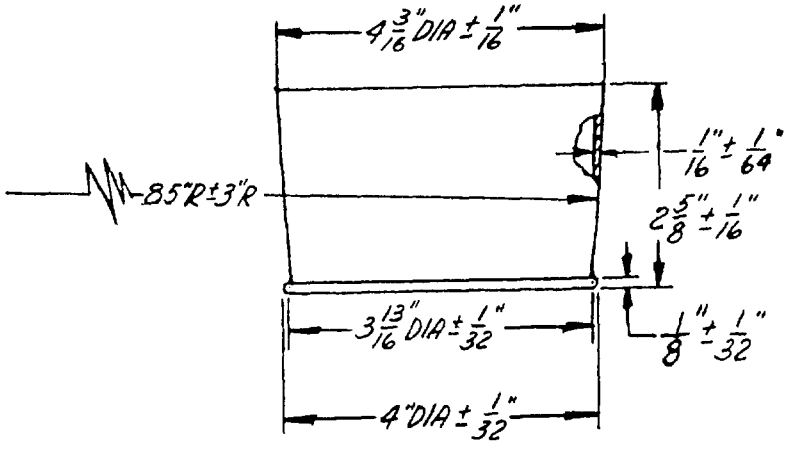


Figure 1. Gloves, Rocket Fuel Handlers; Closure Ring

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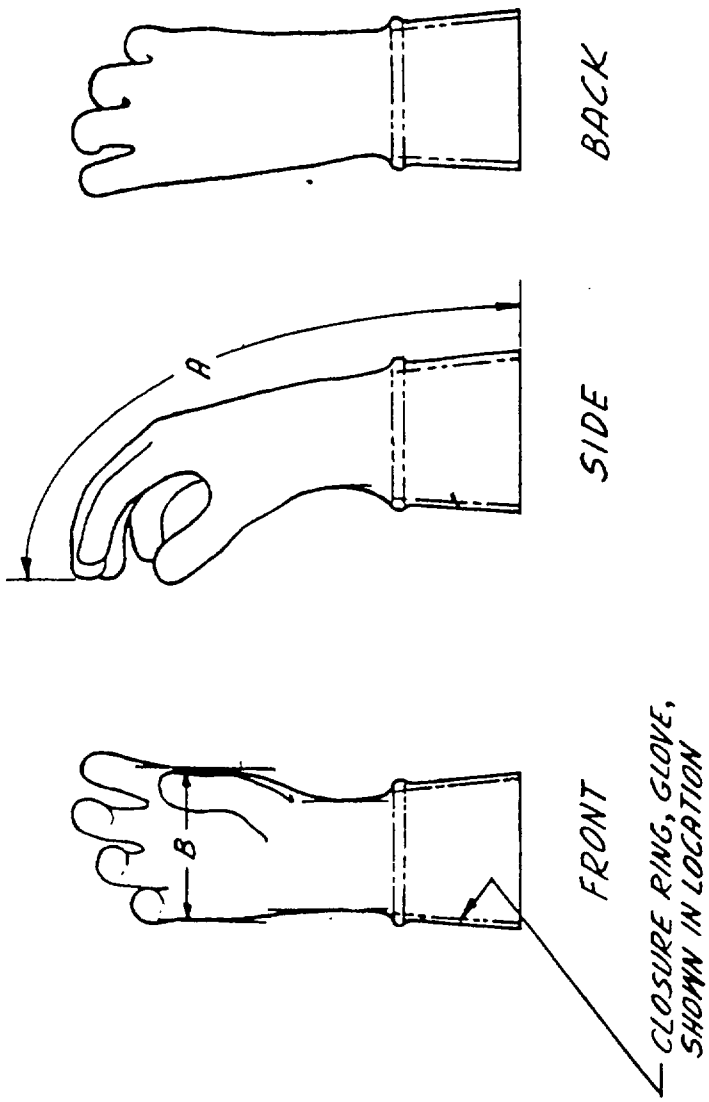


Figure 2. Gloves, Rocket Fuel Handlers

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-G-43196C	Gloves, Rocket Fuel Handlers

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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